

funds could earn substantial profits by buying underpriced convertible bonds and hedging the market risk with short stock positions. While the core strategy remained unchanged, increased competition drove down mispricings and made the sophistication of the convertible bond pricing model more important.

In the simple convertible bond-pricing model, a convertible bond is treated like a corporate bond with a warrant attached. The corporate bond is assumed to provide a floor to how low the convertible price can go. So if you have a 5 percent B grade convertible bond, the assumption is that the price will never be less than a 5 percent nonconvertible B grade bond. We learned that if the stock price fell sufficiently far, then the grade of the bond would also fall, and consequently, the assumed price floor was not a floor. We were able to build that characteristic into the model and get a much more accurate convertible bond-pricing model as a result. Our model also analyzed the entire family of related instruments to the underlying stock, which included convertible bonds, options, warrants, and convertible preferreds, to find optimal hedge combinations across all these related instruments.

After Princeton Newport Partners closed, I called up Fisher Black because I thought we had a better convertible bond model than anyone else and wanted to see if I could sell it to Goldman Sachs. Fisher Black flew out and spent three days reviewing our model. He agreed that it looked very good, but the problem was that the coding of the model was tailored to the Digital Equipment computers we used, which meant they would have had to recode the program to run on their equipment. So he ended up not buying it. The following two or three years, he built his own convertible bond model. I don't know whether he used any of our ideas, but it really doesn't matter.

How important was determining the optimal bet size in your trading success? How and why did you decide to use the Kelly criterion as the method for determining bet size?

The Kelly criterion is the fraction of capital to wager to maximize compounded growth of capital. Even when there is an edge, beyond some threshold, larger bets will result in lower compounded return because of